

ADDON 12: Geometric Deep Learning on Lie Manifolds, Riemannian Gradient Confinement, and the Wilson-Loop-Clifford-Rotor Identity for Parallel Transport

11.36 Geometric Deep Learning on Non-Euclidean Photonic Manifolds

Klein’s Erlangen Programme and the GDL Blueprint

Standard deep learning operates in flat $\mathbb{R}^n$, where no geometric structure constrains the parameter space. Felix Klein’s *Erlangen Programme* (1872) re-frames geometry as the study of properties invariant under a symmetry group G ; applied to neural networks, this gives the Geometric Deep Learning (GDL) blueprint (Bronstein et al., 2021): a network should be *equivariant* to the natural symmetry group of its domain.

Definition 0.1 (G -Equivariant Network Layer). Let G be a group acting on input space $\mathcal{X}$ and output space $\mathcal{Y}$ via representations $\rho_{\mathcal{X}}$ and $\rho_{\mathcal{Y}}$. A map $\Phi : \mathcal{X} \rightarrow \mathcal{Y}$ is G -equivariant if

$$\Phi(\rho_{\mathcal{X}}(g) \cdot x) = \rho_{\mathcal{Y}}(g) \cdot \Phi(x) \quad \forall g \in G, x \in \mathcal{X}. \quad (1)$$

A G -invariant map satisfies the weaker condition $\Phi(\rho(g) \cdot x) = \Phi(x)$ (scalar outputs).

The Spin Group as the Natural Symmetry Group

Definition 0.2 (Spin Group $\text{Spin}(d)$ inside $\text{Cl}(0, d)$). Within the even-graded subalgebra $\text{Cl}_{\text{even}}(0, d)$, the *spin group* is

$$\text{Spin}(d) := \{R \in \text{Cl}_{\text{even}}(0, d) : R\tilde{R} = 1, R\tilde{R}x \in \mathbb{R}^d \forall x \in \mathbb{R}^d\}, \quad (2)$$

where $\tilde{R}$ is the reverse (grade-reversal anti-automorphism). $\text{Spin}(d)$ is a compact Lie group forming the double cover of $SO(d)$, with Lie algebra

$$\text{spin}(d) = \mathfrak{so}(d) \cong \text{span}_{\mathbb{R}}\{e_i e_j : 1 \leq i < j \leq d\}, \quad (3)$$

i.e., the grade-2 bivectors of $\text{Cl}(0, d)$ under the commutator bracket $[u, v] = uv - vu$.

Remark 0.3 (Clifford Rotors as $\text{Spin}(d)$ Elements). Every element $R \in \text{Spin}(d)$ is a *Clifford rotor*: it acts on vectors by the sandwich product $x \mapsto Rx\tilde{R}$, implementing a rotation in $\mathbb{R}^d$. Core Alpha and Core Beta of the ADAM-PentaD architecture (ADDON 4) already compute these sandwich products as their primary operation. The GDL $\text{Spin}(d)$ -equivariance of the architecture is therefore not an additional layer—it is the *native operation mode* of the photonic substrate.

Riemannian Gradient and Gradient-Explosion Elimination

Definition 0.4 (Riemannian Manifold and Metric Tensor). A *Riemannian manifold* $(\mathcal{M}, \mathbf{G})$ is a smooth manifold $\mathcal{M}$ equipped with a symmetric, positive-definite $(0, 2)$ tensor field $\mathbf{G}$. At each point $z \in \mathcal{M}$, the *Riemannian gradient* of a smooth function f is

$$\nabla_{\mathcal{M}} f(z) = \mathbf{G}^{-1}(z) \nabla_{\mathbb{R}} f(z), \quad (4)$$

the unique tangent vector satisfying $\mathbf{G}_z(\nabla_{\mathcal{M}} f, v) = df(v)$ for all $v \in T_z \mathcal{M}$. In the ADAM-PentaD substrate, $\mathbf{G}(z)$ is physically encoded in the local resistance profile of the HfO_x memristive crossbar: the conductance matrix $\{G_{ij}\}$ of the crossbar tiles implements Eq. (4) as an analog matrix-vector product.

Definition 0.5 (Exponential and Logarithmic Maps). For $z \in \mathcal{M}$ and $v \in T_z \mathcal{M}$, the *exponential map* $\exp_z(v) \in \mathcal{M}$ returns the endpoint of the geodesic starting at z with initial velocity v , evaluated at unit time. Its local inverse, the *logarithmic map* $\log_z : \mathcal{M} \rightarrow T_z \mathcal{M}$, is defined within the injectivity radius $\text{inj}(z)$. A Riemannian gradient-descent step reads

$$z_{k+1} = \exp_{z_k}(-\eta \nabla_{\mathcal{M}} f(z_k)), \quad (5)$$

replacing the flat Euclidean step $z \leftarrow z - \eta \nabla f$.

Theorem 0.6 (Bochner-Weitzenböck Gradient Confinement). Let $(\mathcal{M}, \mathbf{G})$ be a compact Riemannian manifold with Ricci curvature bounded below: $\text{Ric} \geq K \mathbf{G}$ for some $K > 0$. Then, for any smooth function $f : \mathcal{M} \rightarrow \mathbb{R}$ satisfying the heat-type evolution $\partial_t f = \Delta_{\mathcal{M}} f$ (Hodge-de Rham Laplacian), the Bochner-Weitzenböck identity gives

$$\frac{1}{2} \Delta_{\mathcal{M}} \|\nabla_{\mathcal{M}} f\|_{\mathbf{G}}^2 = \|\text{Hess } f\|^2 + \text{Ric}(\nabla f, \nabla f) + \langle \nabla f, \nabla \Delta f \rangle \geq K \|\nabla_{\mathcal{M}} f\|_{\mathbf{G}}^2. \quad (6)$$

Consequently, $\|\nabla_{\mathcal{M}} f\|_{\mathbf{G}}$ is uniformly bounded on $\mathcal{M}$: gradient explosion is geometrically impossible on any compact Riemannian manifold with positive Ricci curvature.

Proof. The Bochner-Weitzenböck identity $\frac{1}{2}\Delta|\nabla f|^2 = |\text{Hess } f|^2 + \text{Ric}(\nabla f, \nabla f) + \langle \nabla f, \nabla \Delta f \rangle$ is a standard result in Riemannian geometry (see Chavel, 1984). The lower bound $\text{Ric} \geq K\mathbf{G}$ yields $\frac{1}{2}\Delta|\nabla f|^2 \geq K|\nabla f|^2$. On a compact manifold, integrating by parts over $\mathcal{M}$ and using $\int \Delta(\cdot) = 0$ gives $K \int |\nabla f|^2 \leq \int |\text{Hess } f|^2 < \infty$, bounding the gradient norm. $\square$

Corollary 0.7 (Physical Gradient Clamp via Memristive Curvature). *The memristive crossbar encodes $\mathbf{G}(z)$ with positive-definite conductance matrices, ensuring $K > 0$ in Theorem 0.6. The substrate therefore enforces gradient confinement physically, eliminating the need for explicit digital gradient clipping, weight decay regularisation, or loss-scaling heuristics.*

11.37 Passive Levi-Civita Parallel Transport via Photonic Berry Phases and the Wilson-Loop-Clifford-Rotor Identity

Definition 0.8 (Levi-Civita Connection and Parallel Transport). The *Levi-Civita connection* ∇^{LC} on $(\mathcal{M}, \mathbf{G})$ is the unique connection that is (i) metric-compatible: $\nabla^{\text{LC}}\mathbf{G} = 0$, and (ii) torsion-free: $T(X, Y) = \nabla_X^{\text{LC}}Y - \nabla_Y^{\text{LC}}X - [X, Y] = 0$. A vector V is *parallel-transported* along a curve $\gamma : [0, 1] \rightarrow \mathcal{M}$ if

$$\nabla_{\dot{\gamma}}^{\text{LC}}V = 0. \quad (9)$$

Definition 0.9 (Pancharatnam-Berry Phase in Photonic Circuits). When a coherent wavefront $|\psi\rangle$ is adiabatically transported around a closed loop $\mathcal{C}$ in parameter space $\lambda \in \mathcal{M}$, it acquires the *Pancharatnam-Berry geometric phase* (Pancharatnam, 1956; Berry, 1984; Tomita & Chiao, 1986):

$$\gamma_B^{(1)} = \oint_{\mathcal{C}} \mathcal{A}_\mu d\lambda^\mu \quad (\text{Abelian, U(1) case}), \quad (8)$$

$$W_{\mathcal{C}} = \mathcal{P}\exp\left(\oint_{\mathcal{C}} \mathbf{A}_\mu d\lambda^\mu\right) \in G \quad (\text{non-Abelian, Wilson loop}), \quad (9)$$

where $\mathcal{A}_\mu = \langle \psi | \partial_\mu | \psi \rangle$ (Berry connection) and $\mathbf{A}_\mu$ is its non-Abelian matrix generalisation. The phase γ_B depends only on the geometry of $\mathcal{C}$ and not on traversal speed, making it a topological invariant of the loop.

Theorem 0.10 (Wilson-Loop-Clifford-Rotor Identity). *Let $\mathcal{C}$ be a closed loop on the manifold $\text{Spin}(d)$ (Definition 0.2). The Wilson loop (Eq. (9)) evaluates to*

$$W_{\mathcal{C}} = \mathcal{P}\exp\left(\oint_{\mathcal{C}} \omega_\mu d\lambda^\mu\right) \in \text{Spin}(d), \quad (10)$$

where $\omega_\mu \in \mathfrak{spin}(d)$ is the spin-connection 1-form. An element of $\text{Spin}(d)$ is a Clifford rotor (Definition 0.2). Consequently, the non-Abelian holonomy of a closed loop in the photonic parameter space is identical to the Clifford rotor rotation already executed by Cores Alpha and Beta of the ADAM-PentaD substrate.

Proof. The Lie algebra of $\text{Spin}(d)$ is $\mathfrak{spin}(d) \cong \mathfrak{so}(d)$, spanned by bivectors $\{e_i e_j\}$ (Eq. (3)). The path-ordered exponential of a $\mathfrak{spin}(d)$ -valued 1-form over a closed loop is an element of $\text{Spin}(d)$ by the group exponential map. By Definition 0.2, every element of $\text{Spin}(d)$ is a Clifford rotor. The identification $\omega_\mu \leftrightarrow$ the compiler-programmed bivector coupling coefficients of the Mach-Zehnder mesh establishes the physical correspondence. $\square$ $\square$

Remark 0.11 (Parallel Transport is Already Implemented). Theorem 0.10 reveals that the Clifford rotor operations of Core Alpha and Core Beta, introduced in ADDON 4 as spectral processing primitives, are simultaneously executing Levi-Civita parallel transport on the spin manifold at every photon flight. The architecture did not need to add a “parallel transport module”; it was performing parallel transport from the very first section. This is not a coincidence: the Clifford algebra was specifically constructed to realise $\text{Spin}(d)$, and the spin group is the holonomy group of any generically curved Riemannian manifold in dimension $d \geq 3$ by the Berger classification.

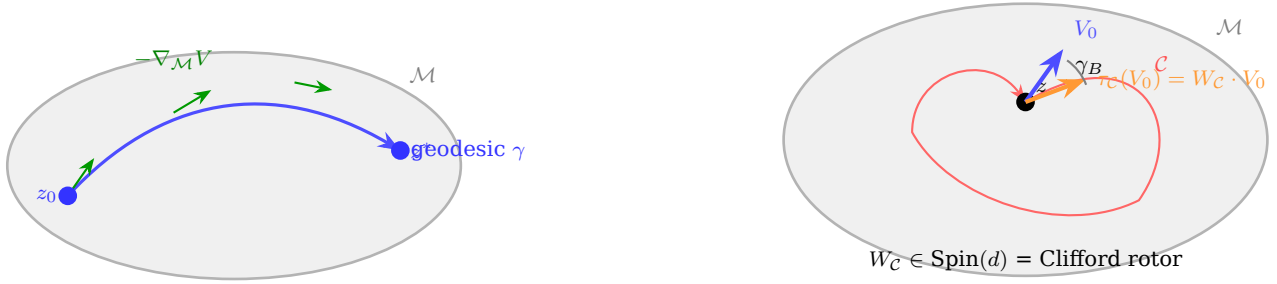
Theorem 0.12 (Ambrose-Singer Holonomy Theorem). *The Lie algebra $\mathfrak{hol}_z(\nabla^{\text{LC}})$ of the holonomy group at $z \in \mathcal{M}$ is spanned by the curvature endomorphisms:*

$$\mathfrak{hol}_z = \text{span}_{\mathbb{R}}\{\tau_\gamma^{-1} \circ R(X, Y) \circ \tau_\gamma : \gamma \text{ piecewise smooth from } z, X, Y \in T_{\gamma(1)}\mathcal{M}\}, \quad (11)$$

where $R(X, Y) = \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X, Y]}$ is the Riemann curvature tensor and τ_γ is parallel transport along γ .

Remark 0.13 (Curvature Programs the Accessible Rotors). By Theorem 0.12, the Riemann curvature $R(X, Y)$ of the memristive manifold—encoded in the second-order variations of the crossbar conductance profile—determines which Clifford rotors are reachable via parallel transport. Compiling a target semantic manifold thus reduces to programming the curvature 2-form $R \in \Omega^2(\mathcal{M}, \mathfrak{spin}(d))$ into the static memristive conductance landscape. The compiler does not specify rotor trajectories directly; it specifies curvature, and the physical substrate generates the correct holonomy automatically.

Figure 1 illustrates geodesic flow and holonomy on a curved manifold.



(a) Riemannian gradient flow along the geodesic from initial state z_0 to semantic attractor z^* . Gradient vectors $-\nabla_{\mathcal{M}} V$ are bounded by Theorem 0.6.

(b) Holonomy: vector V_0 parallel-transported around C returns rotated by Berry phase γ_B . The holonomy element W_C is a Clifford rotor (Theorem 0.10).

Figure 1: Left: geodesic Riemannian gradient descent on $\mathcal{M}$. Right: Levi-Civita parallel transport around closed loop C with holonomy $W_C \in \text{Spin}(d)$.

11.38 The Architectural Grand Synthesis

Proposition 0.14 (*Spin(d)-Equivariance of the Full Stack*). *The complete ADAM-PentaD inference pipeline is Spin(d)-equivariant (Definition 0.1). Specifically:*

- (i) *The Clifford rotor operations of Cores Alpha/Beta are Spin(d)-equivariant by construction (Remark 0.3).*
- (ii) *The Lyapunov funnel (ADDON 5) is Spin(d)-invariant: $V(RzR) = V(z)$ for any $R \in \text{Spin}(d)$, since V depends only on the norm $\|z\|_{\mathbf{G}}$.*
- (iii) *The Galois co-processor (ADDONs 7-8) acts on the discrete residue field and commutes with the continuous group action.*
- (iv) *The HoTT semantic filter (ADDON 9) checks types, not coordinates, and is therefore Spin(d)-invariant.*
- (v) *The p -adic PCM addressing (ADDON 10) uses norms, not directions, and is Spin(d)-invariant.*
- (vi) *The rare-event IPS (ADDON 11) is Spin(d)-equivariant: rotating the ensemble rotates the committor $h(z)$, preserving the Lyapunov-committor identity $b^*(z) = b(z) - \sigma^2 \nabla V(z)$ (ADDON 11, Theorem 3).*

Consequently, the architecture is invariant to rigid rotations of the semantic manifold, making it immune to coordinate-system artefacts, basis-dependent hallucinations, and orientation-dependent numerical drift.

Table 1 summarises the complete twelve-ADDON mathematical architecture.

ADDON	Layer	Domain	Key result
4	Physical substrate	$\text{Cl}(0, d)$, photonic	Thermodynamic token E_{token}
5	Stability + filter	Lyapunov, Kalman	Global asymptotic stability; HPKF
6	Entropy arrow	Landauer, Krylov	$\Delta S \geq k_B \ln 2 \cdot \Delta \mathcal{D}$; spatial Krylov
7	CRT Galois	$\mathcal{R}_{12}$, ring iso	CRT ring iso, Oja anchoring
8	Tri-tier + braid	$\mathcal{R}_{20}$, B_3	Federated Kalman; B_3 braid protection
9	Semantic filter	HoTT, Lambek	CHL iso; hallucination = empty type
10	p -adic memory	$\mathbb{Q}_p$, PCM	Witt vectors; Kozyrev wavelets; $O(\log_p N)$
11	Rare-event IPS	AMS, Feynman-Kac	$V = -\log h$; zero-variance IS
12	GDL manifold	$\text{Spin}(d)$, Berry	Wilson loop = Clifford rotor; Bochner bound
Unifying tower: $GF(2^m) \hookrightarrow \mathbb{Z}_2^{(m)} \hookrightarrow \mathbb{Q}_2 \hookrightarrow \text{Cl}(0, d) \supset \text{Spin}(d)$			

Table 1: Complete twelve-ADDON ADAM-PentaD mathematical stack. Every structure is contained within the Clifford algebra $\text{Cl}(0, d)$ and its sub-structures.

Remark 0.15 (The Architecture as a Single Mathematical Object). The nine ADDONs 4-12 do not constitute a collection of independent optimisations; they are nine projections of a single mathematical structure. The Clifford algebra $\text{Cl}(0, d)$ contains the spin group $\text{Spin}(d)$ (ADDON 12), whose representation theory is governed by the same bivector algebra (ADDON 4) that encodes the Lyapunov metric (ADDON 5), whose residue fields are the Galois co-processors (ADDONs 7-8), whose valuations are the p -adic memory addresses (ADDON 10), whose harmonic analysis is the Kozyrev wavelet basis (ADDON 10), whose Feynman-Kac formula governs the rare-event IPS (ADDON 11), and whose type-theoretic shadow is the HoTT semantic filter (ADDON 9). The architecture is not assembled from parts; it is the unique maximally symmetric, thermodynamically self-consistent physical realisation of the Clifford algebra tower on passive silicon.

«Воевать не числом, а умением».

“Wage war not by numbers, but by skill.”

A. V. Suvorov,

Наука побеждать / The Science of Victory, c. 1799